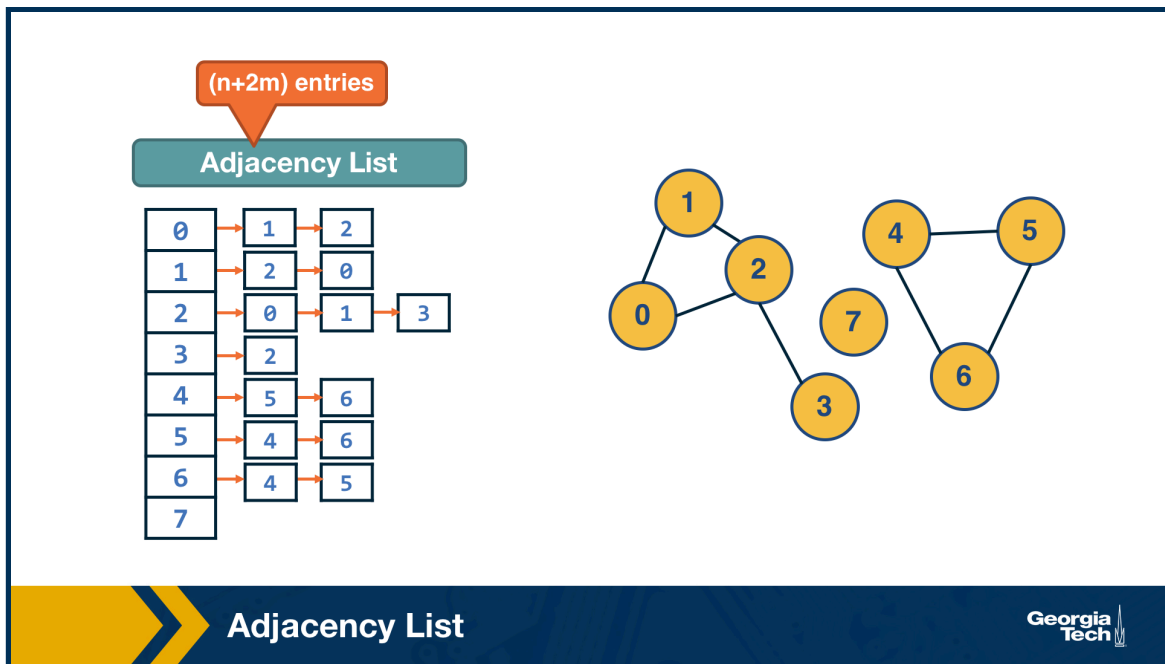


L2: Adjacency List



The adjacency list representation requires $n+2*m$ space because every edge is included twice.

The difference between adjacency matrices and lists can be very large when the graph is **sparse**. A graph is sparse if the number of edges m is much closer to the number of nodes n than to the maximum number of edges ($n\text{-choose-}2$). In other words, the adjacency matrix of a sparse graph is mostly zeros.

A graph is referred to as **dense**, on the other hand, if the number of edges is much closer to $n\text{-choose-}2$ than to n .

It should be noted that most real-world networks are sparse. The reason may be that in most technological, biological and social networks, there is a cost associated with each edge – dense networks would be more costly to construct and maintain.

Food for Thought

Suppose that a network grows by one node in each time unit. The new node always connects to k existing nodes, where k is a constant. As this network grows, will it gradually become sparse or dense (*when n becomes much larger than k*)?